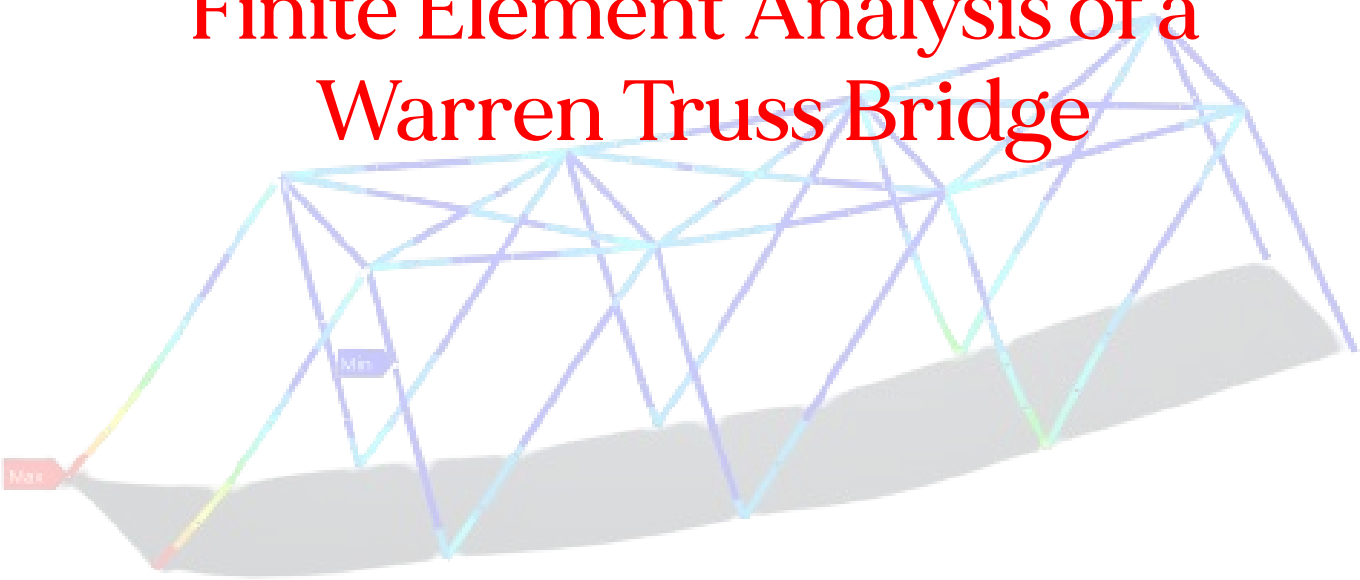
INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

Department of Civil Engineering

Structural Engineering II (CE 302) Project

Finite Element Analysis of a Warren Truss Bridge



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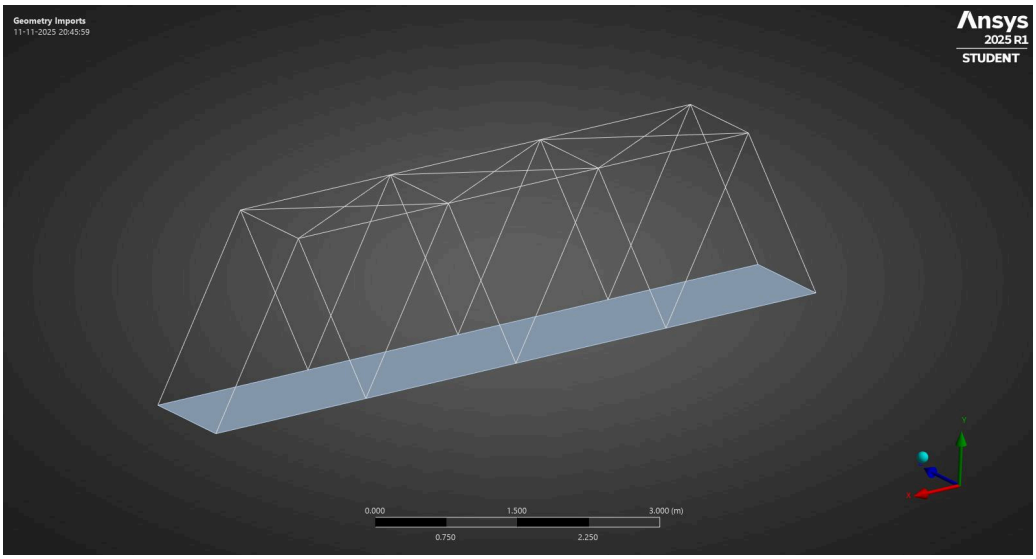
Ansys

Abstract

This report presents a static structural finite element analysis of an 8-meter span Warren truss bridge modeled in ANSYS Mechanical. The structure was designed with structural steel members of 0.05 m × 0.05 m square cross-section and included a deck modeled as a shell element. A uniform pressure load of 0.2 Pa was applied normal to the deck surface to simulate distributed loading. The analysis focused on determining axial forces, direct and equivalent stresses, bending moments, shear forces, and total deformation. The maximum von-Mises stress obtained was 59.27 Pa, and the maximum total deformation was 6.61×10^{-9} m, indicating negligible structural response under the applied loading condition.



1. Introduction

Truss bridges are structural systems composed of interconnected members forming triangular units, ensuring stability and efficient load distribution. The Warren truss, characterized by equilateral triangles, provides uniform force distribution and material efficiency.

This analysis aims to evaluate the structural performance of a Warren truss bridge under a small distributed pressure load using Finite Element Analysis (FEA) in ANSYS Mechanical. The focus is on determining deformation patterns, stress concentrations, and internal forces within the bridge components.

2. Geometry and Design Data

- **Bridge Type:** Warren Truss
- **Span Length:** 8.0 m
- **Height of Truss:** 2.0 m
- **Member Cross-Section:** 0.05 m × 0.05 m (square hollow section)
- **Deck**: Modeled as a shell element spanning the lower chords of the truss
- **Material:** Structural Steel:
  - Young's Modulus (E): 210 GPa
  - Poisson's Ratio (v): 0.3
  - Density: 7850 kg/m³

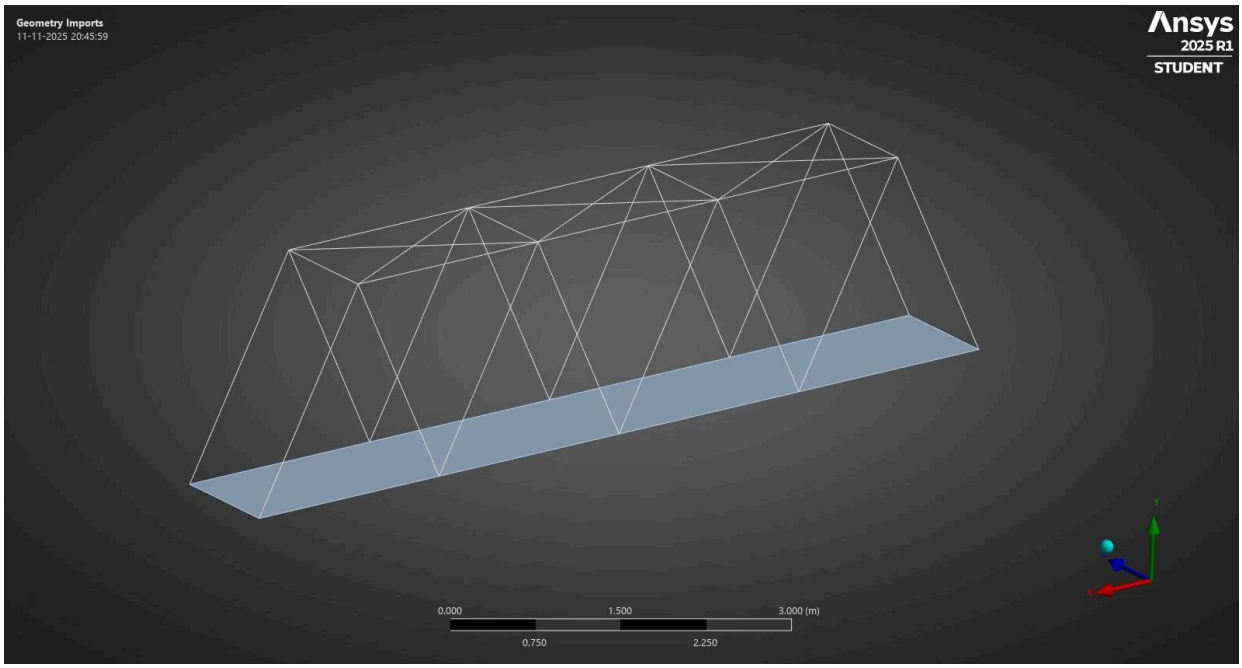


Figure 1: Bridge Geometry

3. Meshing

The bridge was discretized using a combination of beam elements for truss members and shell elements for the deck surface.

Mesh Parameters:

Parameter	Value
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	0.35355 m
Growth Rate	1.2
Adaptive Sizing	Disabled
Minimum Edge Length	1.0 m
Mesh Defeaturing	Enabled (1.7678e-003 m)
Smoothing	Medium
Mesh Quality	No Errors (Aggressive Mechanical Limit)

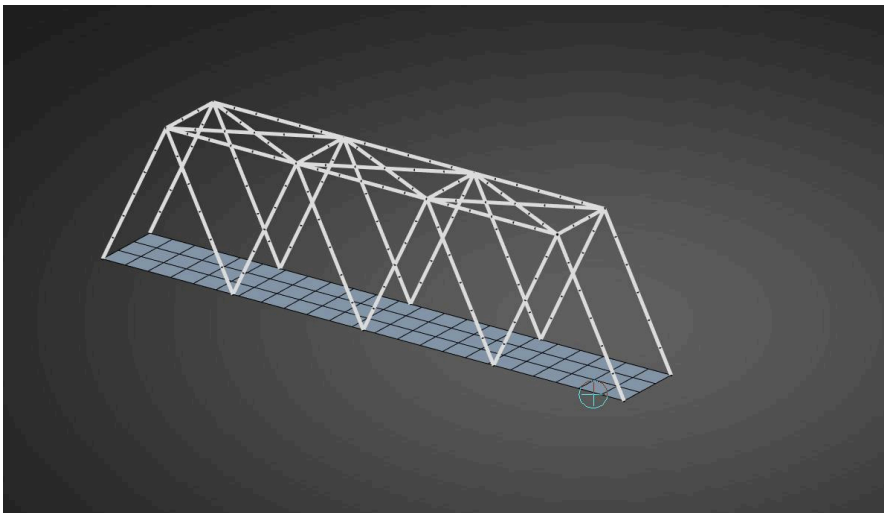


Figure 2: Meshed Model

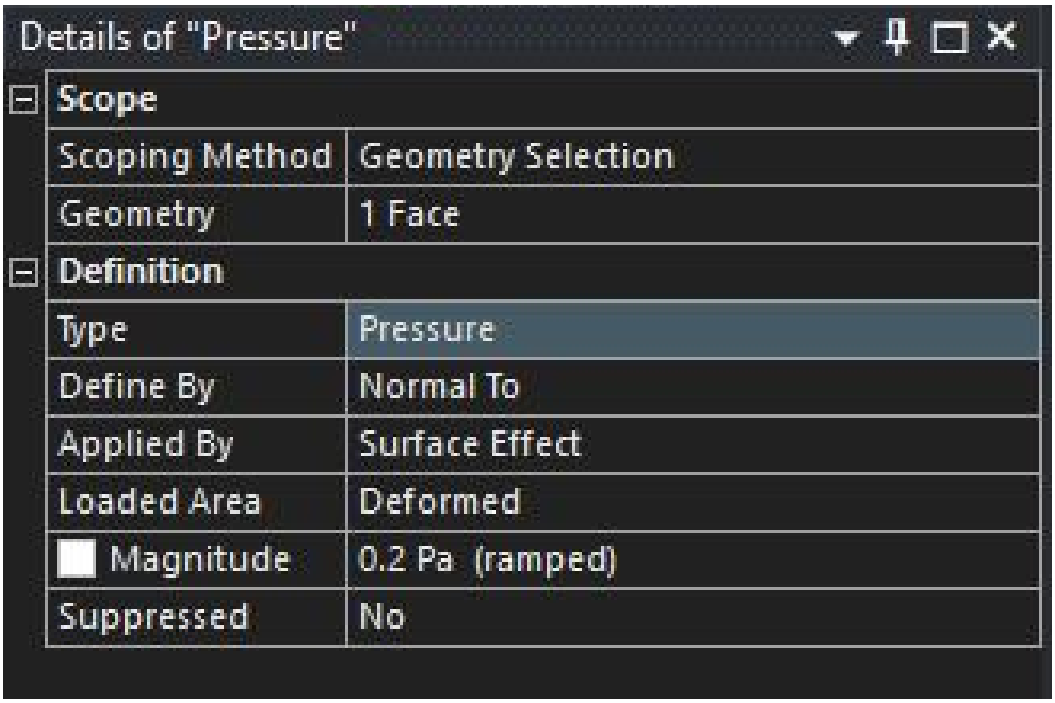
4. Boundary Conditions and Loading

Boundary Conditions:

- Left End: Displacement Support (translations restrained)
- Right End: Fixed Support (all degrees of freedom restrained)

Loading Conditions:

- Type: Surface Pressure
- Magnitude: 0.2 Pa (ramped)
- Direction: Normal to deck surface
- Scope: Applied on one deck face



Details of "Pressure"

Scope	
Scoping Method	Geometry Selection
Geometry	1 Face
Definition	
Type	Pressure
Define By	Normal To
Applied By	Surface Effect
Loaded Area	Deformed
Magnitude	0.2 Pa (ramped)
Suppressed	No

Figure 3: Loading Condition

5. Analysis Settings

- **Analysis Type:** Static Structural
- **Solver Type:** Program Controlled (Static)
- **Material Behavior:** Linear Elastic, Isotropic
- **Units:** SI (m, N, Pa)
- **Solution Output:** Axial Force, Direct Stress, Equivalent (von-Mises) Stress, Total Bending Moment, Total Deformation, and Shear–Moment Diagrams

6. Results

6.1 Summary of Numerical Results

Parameter	Maximum	Minimum	Units
Axial Force (X-Direction)	+0.34359	−0.34424	N
Direct Stress (Top/Bottom)	+137.44	−137.70	Pa
Equivalent (von-Mises) Stress	59.274	2.409	Pa
Total Bending Moment	7.1457×10^{-4}	2.5738×10^{-6}	N-m
Total Deformation	6.6077×10^{-9}	0	m
Shear Force (Deck)	4.7579×10^{-4}	2.88×10^{-10}	N
Bending Moment (Deck)	3.4965×10^{-4}	1.4003×10^{-5}	N-m
Displacement (Deck)	5.8838×10^{-9}	1.0981×10^{-10}	m

6.2 Contour Plots and Diagrams

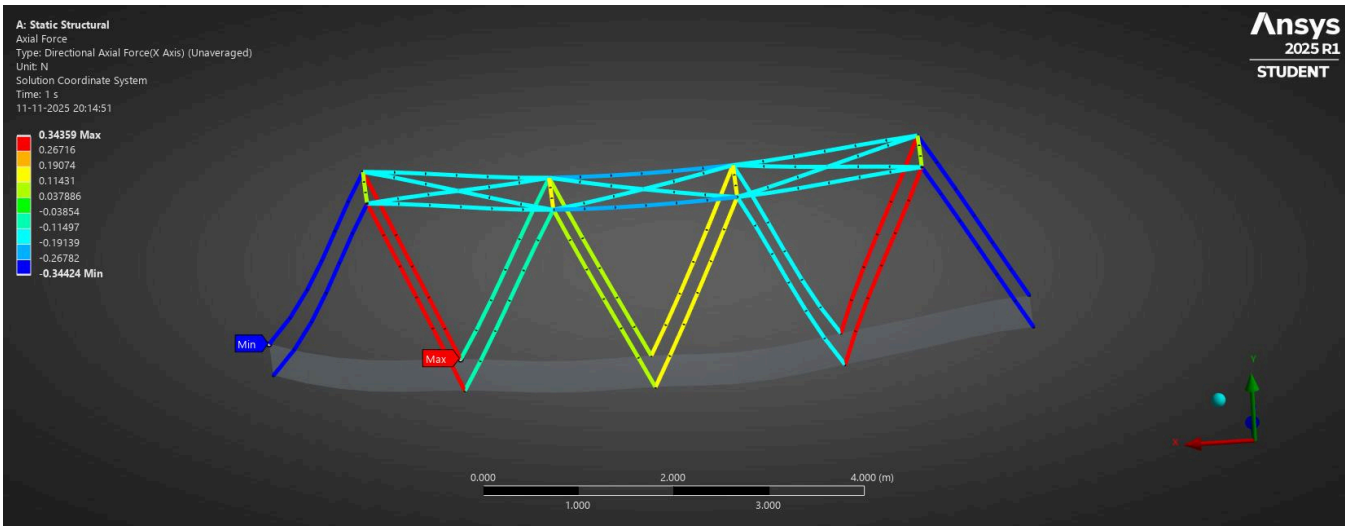


Figure 4: Axial Force Distribution

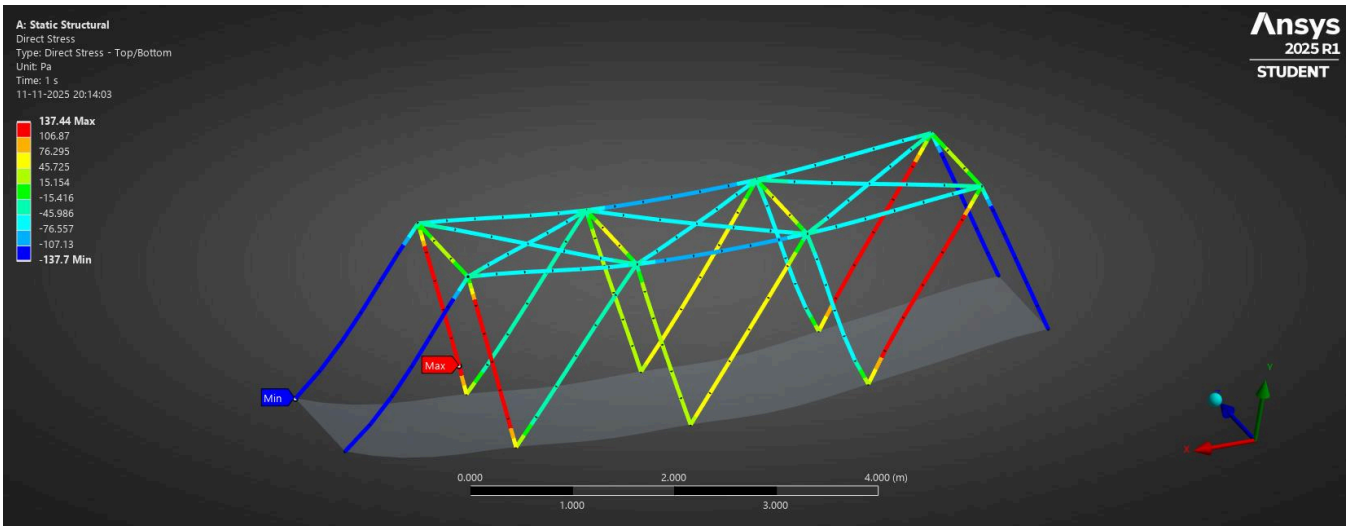


Figure 5: Direct Stress Distribution

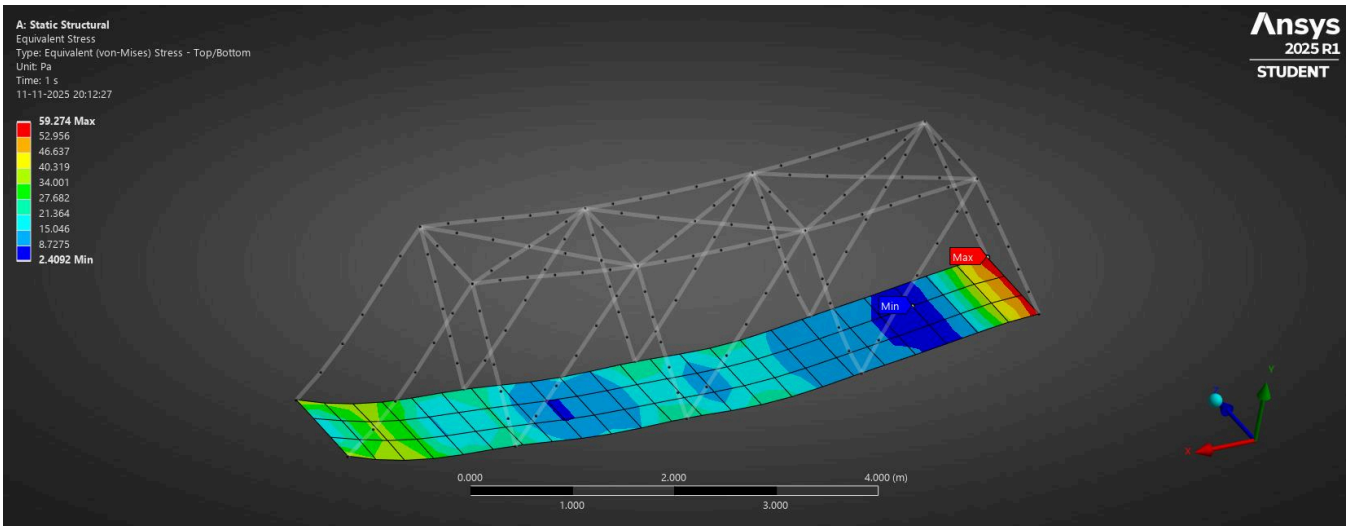


Figure 6: Equivalent (von-Mises) Stress

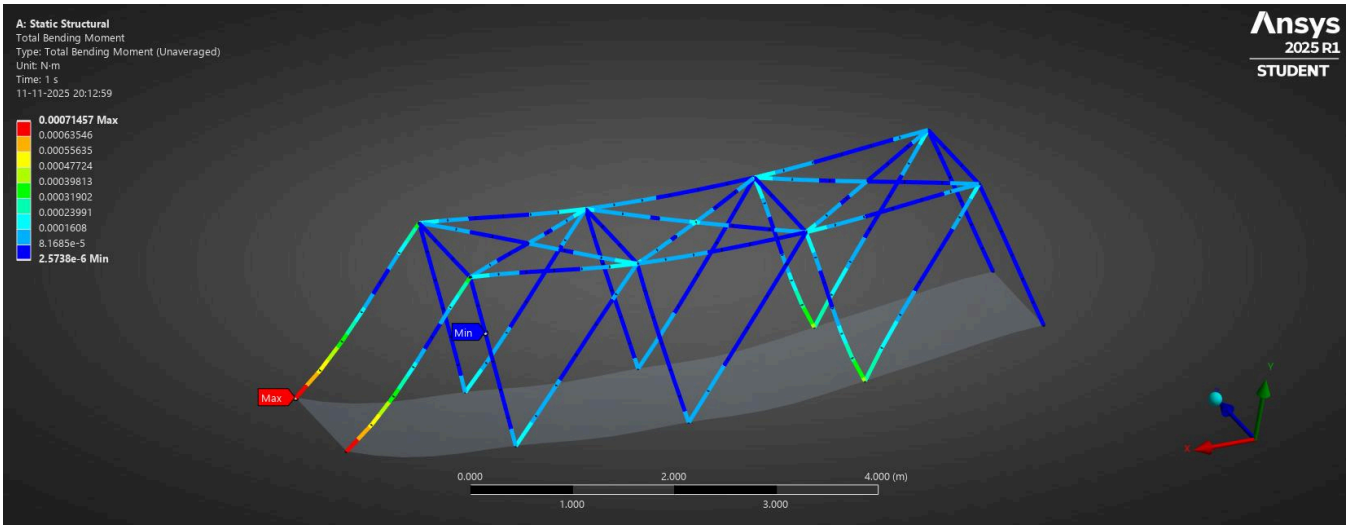


Figure 7: Total Bending Moment

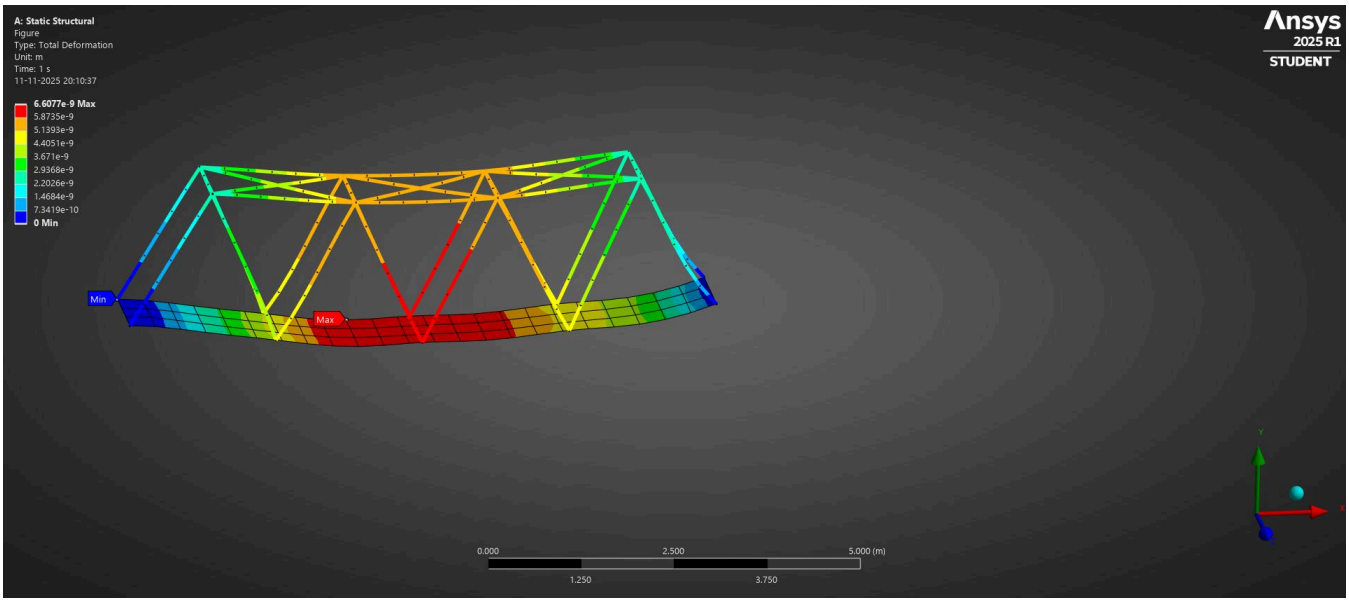


Figure 8: Total Deformation

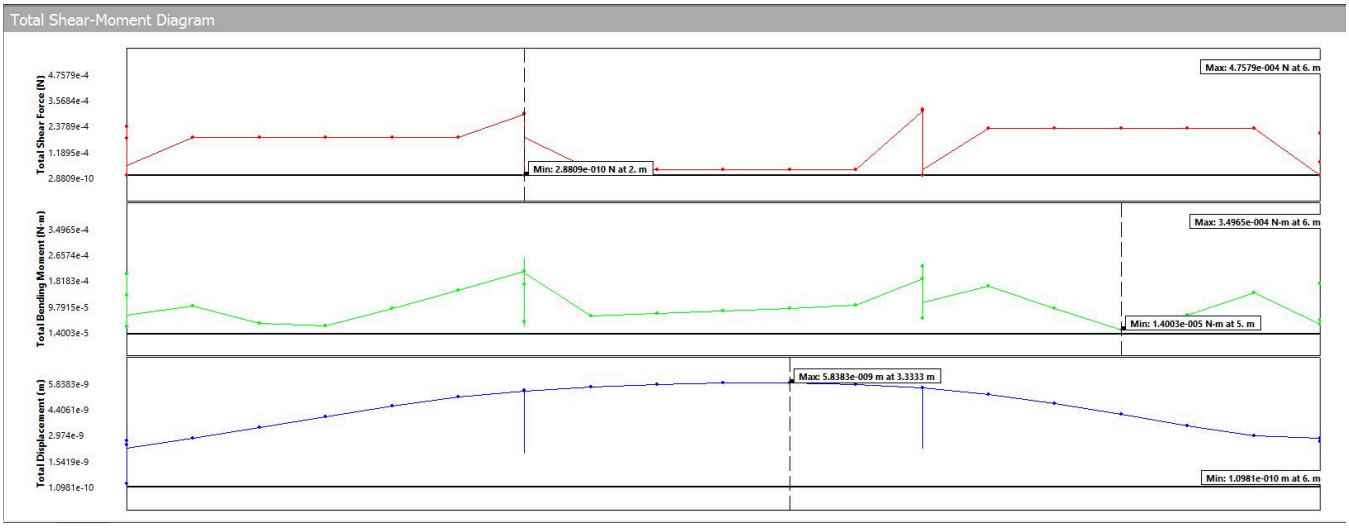


Figure 9: Shear–Moment Diagram

7. Discussion of Results

- The Warren truss exhibited the expected behavior, with alternating tension and compression in diagonal members visible in the axial force contours.
- The maximum direct stress observed was 137.44 Pa, occurring near the top compression members close to the supports.-
- Equivalent (von-Mises) stress distribution showed a maximum of 59.27 Pa, concentrated on the deck near the loaded region.
- Bending moments in the truss members were very small (7.14×10^{-4} N·m), confirming that the bridge primarily carries loads through axial forces.
- The total deformation of the structure (6.61×10^{-9} m) is negligible for the applied load, indicating an extremely stiff system relative to the load magnitude.
- Shear and bending diagrams of the deck show a maximum shear of 4.7579×10^{-4} N and bending moment of 3.4965×10^{-4} N·m, both occurring near mid-span, consistent with symmetrical loading.

8. Conclusion

- The finite element analysis of the Warren truss bridge under a uniform pressure of 0.2 Pa produced very small stresses and deformations.
  - Maximum von-Mises Stress: 59.27 Pa
  - Maximum Total Deformation: 6.61×10^{-9} m
- The stress magnitudes are negligible compared to the material yield strength of structural steel (≈250 MPa), indicating a structurally safe condition for this loading scenario.
- The analysis verified that the Warren truss effectively distributes loads through axial action with minimal bending in members. The results also confirm that the modeling and boundary conditions were correctly implemented.

9. Scope of Improvement

- For realistic loading conditions, higher magnitude loads (in kN/m^2) should be applied to simulate vehicular or pedestrian loads.
- Reaction forces should be evaluated to confirm static equilibrium.
- A mesh convergence study can be performed for refined accuracy.
- Dynamic or fatigue analysis can be conducted for long-term performance evaluation under cyclic loading.
- Multiple load cases such as point loading and asymmetric loads should be examined for worst-case behavior.